

Standard Operating Procedure

Title: KTF Service Team Preventative Maintenance Form

Doc. No: IF.KTF-POP.1146

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Customer Information			
Customer Company Name:	SILQ	Customer Address	311 Otterson Dr. #60 Chico, Ca. 95928
P.O. Number	0000066	Bldg. / room #	
Contact Name	Chris Turner	Contact Phone	661-860-9704
Contact E-mail	christ@silq.tech	Customer Asset #	ST-005
Repligen FSE	Jeff Parker		

(7.1) Preventive Maintenance System Component Verification and Replacement				
Component	Type	Serial / Lot number	Needs to be Replaced?	Communication Verification
System	KTF 2000	SN# 190603-001	No	(Yes) No / N/A
Feed Pump 1	Levitronix	SN# 160919-0710	No	(Yes) No / N/A
Feed Pump 2	Levitronix	SN# 160919-0705	No	(Yes) No / N/A
Buffer Pump 1	Masterflex	SN# ①	No	(Yes) No / N/A
Buffer Pump Head 1	Spectrum	SN# A20001737	N/A	N/A
Buffer Pump 2	Masterflex	SN# ①	No	(Yes) No / N/A
Buffer Pump Head 2	Spectrum	SN# A20001748	N/A	N/A
Permeate Pump 1		SN#		Yes / No / N/A
Permeate Pump Head 1		SN#	N/A	N/A
Permeate Pump 2		SN# N/A		Yes / No / N/A
Permeate Pump Head 2		SN#	N/A	N/A
Product Pump		SN#		Yes / No / N/A
Product Pump Head		SN#	N/A	N/A
Retentate Flowmeter 1	Spectrum	SN# 10026	No	(Yes) No / N/A
Retentate Flowmeter 2	Spectrum	SN# 10045	No	(Yes) No / N/A
Permeate Flowmeter 1	Spectrum	SN# 10009	No	(Yes) No / N/A
Permeate Flowmeter 2		SN#		Yes / No / N/A
Feed Flowmeter 1		SN#		Yes / No / N/A

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Feed Flowmeter 2		SN#	N/A	SP 01 Nov 2022	Yes / No / N/A
Back Pressure Control Valve 1	BPCV-01	SN#	N/A	No	☒ Yes / No / N/A
Back Pressure Control Valve 2	BPCV-02	SN#	N/A	No	☒ Yes / No / N/A
HMI	Allen Bradley	SN#	(21) 6F80I9KQ	No	☒ Yes / No / N/A
Pressure Sensor (1) Feed		SN#			Yes / No / N/A
Pressure Sensor (2) Feed		SN#			Yes / No / N/A
Pressure Sensor (3) Retentate		SN#			Yes / No / N/A
Pressure Sensor (4) Retentate		SN#			Yes / No / N/A
Pressure Sensor (5) Permeate		SN#			Yes / No / N/A
Pressure Sensor (6) Permeate		SN#			Yes / No / N/A
Scale 1		SN#			Yes / No / N/A
Scale 2		SN#			Yes / No / N/A
Scale 3		SN#			Yes / No / N/A
Touch Panel Calibration		N/A			N/A
Comments:	SP 01 Nov 2022				

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(7.2) Pressure Sensor Accuracy Test

Pressure test with calibrated pressure gauge.

Note: Acceptance criteria per POP-SOP-1146 Section 7.2

As Found Calibration Test

Initial Calibration	0 PSI or 0 mBar		15 PSI or 1034 mBar		30 PSI or 2068 mBar		Pass or Fail	
	Ref. Gauge	HMI	Ref. Gauge	HMI	Ref. Gauge	HMI	Pass	Fail
P _{fec}	0.00	0.00	15.00	15.05	30.00	30.11	✓	
P _{ret}	0.00	0.00	15.00	15.05	30.00	30.13	✓	
P _{perm}	0.00	0.00	15.00	15.06	30.00	30.11	✓	

Comments:

PE-04 0.00/0.00 15.00/15.03 30.00/30.05

PE-05 0.00/0.00 15.00/15.08 30.00/30.11

Post Pressure Calibration (if as found calibration test failed)

Initial Calibration	0 PSI or 0 mBar		15 PSI or 1034 mBar		30 PSI or 2068 mBar		Pass or Fail	
	Ref. Gauge	HMI	Ref. Gauge	HMI	Ref. Gauge	HMI	Pass	Fail
P _{fec}								
P _{ret}								
P _{perm}								

Comments:

N/A SP or New 2022

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(7.3) Pump RPM Test
Note: Acceptance criteria per POP-SOP-1146 Section 7.3

Recirculating Pump 1		Recirculating Pump 2		Buffer Pump 1		Buffer Pump 2		Permeate Pump 1		Permeate Pump 2		Product Pump	
Max Pump Speed		Max Pump Speed		Max Pump Speed		Max Pump Speed		Max Pump Speed		Max Pump Speed		Max Pump Speed	
4000		4000		640		1400							
Set RPM	Actual RPM	Set RPM	Actual RPM	Set RPM	Actual RPM	Set RPM	Actual RPM	Set RPM	Actual RPM	Set RPM	Actual RPM	Set RPM	Actual RPM
25%	1000	25%	600	25%	159.8	25%	159.4	25%		25%		25%	
50%	2000	50%	2000	50%	320.0	50%	319.7	50%		50%		50%	
75%	3000	75%	3000	75%	480.6	75%	480.2	75%		75%		75%	
100%	4000	100%	4000	100%	640.1	100%	640.2	100%		100%		100%	
Pass or Fail or N/A		Pass or Fail or N/A		Pass or Fail or N/A		Pass or Fail or N/A		Pass or Fail or N/A		Pass or Fail or N/A		Pass or Fail or N/A	

Comments:

- ① Entry Error 640 JP 01 Nov 2022
- ② Entry Error 640.0 JP 01 Nov 2022

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(7.4) Flowmeter Test (100/600/1000)

Note: Acceptance criteria per POP-SOP-1146 Section 7.4

Flowmeter path:		
Flowmeter Tag #	Flowmeter Type	
FL-01	Spectrum	
HMI Flow Rate Reading	Reference Sensor Flow Rate	Status Pass / Fail
9.8	9.82	Pass / Fail / NA
20.8	20.75	Pass / Fail / NA
42.1	42.56	Pass / Fail / NA
Flowmeter path:		
Flowmeter Tag #	Flowmeter Type	
FL-03	Spectrum	
HMI Flow Rate Reading	Reference Sensor Flow Rate	Status Pass / Fail
1.1	1.09	Pass / Fail / NA
2.2	2.21	Pass / Fail / NA
4.1	4.14	Pass / Fail / NA
Flowmeter path:		
Flowmeter Tag #	Flowmeter Type	
FL-02	Spectrum	
HMI Flow Rate Reading	Reference Sensor Flow Rate	Status Pass / Fail
10.9	11.11	Pass / Fail / NA
21.1	21.53	Pass / Fail / NA
39.9	40.20	Pass / Fail / NA

Comments:

N/A JP 01 Nov 2022

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(7.4) Flowmeter Test (100/600/1000)

Note: Acceptance criteria per POP-SOP-1146 Section 7.4

Flowmeter path:

Flowmeter Tag #	Flowmeter Type	Status
		Pass / Fail
HMI Flow Rate Reading	Reference Sensor Flow Rate	Pass / Fail / NA
		Pass / Fail / NA
		Pass / Fail / NA

Flowmeter path:

Flowmeter Tag #	Flowmeter Type	Status
		Pass / Fail
HMI Flow Rate Reading	Reference Sensor Flow Rate	Pass / Fail / NA
		Pass / Fail / NA
		Pass / Fail / NA

Flowmeter path:

Flowmeter Tag #	Flowmeter Type	Status
		Pass / Fail
HMI Flow Rate Reading	Reference Sensor Flow Rate	Pass / Fail / NA
		Pass / Fail / NA
		Pass / Fail / NA

Comments:

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(7.5) Scale Input Verification Test

Note: Acceptance criteria per POP-SOP-1146 Section 7.5

Customer is responsible for annual in-house calibration

Scale	Test 1		Test 2		Test 3		Pass, Fail, (N/A)
Circle type	4mA or Ref. Weight	HMI Weight Reading	12 mA or Ref. Weight	HMI Weight Reading	20 mA or Ref. Weight	HMI Weight Reading	
WE-01	4mA	0.00	12mA	226.8	20mA	453.6	Pass
Circle type	4mA or Reference Weight	HMI Weight Reading	12 mA or Ref. Weight	HMI Weight Reading	20 mA or Ref. Weight	HMI Weight Reading	
WE-02							
Circle type	4mA or Reference Weight	HMI Weight Reading	12 mA or Ref. Weight	HMI Weight Reading	20 mA or Ref. Weight	HMI Weight Reading	
WE-03							

Comments:

N/A SP 01 Nov 2022

Controller Serial Number:

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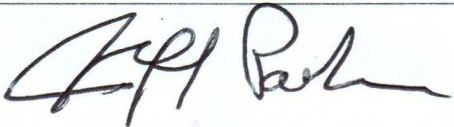
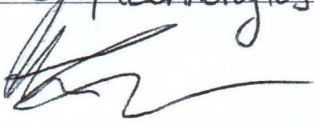
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(11) Preventative Maintenance Verification

System ready for release?

☒ Yes ☐ No

(12) Customer Acceptance and Sign Off

Authorized Repligen Service Engineer		Customer	
Name (print)	Jeff Parker	Name (print)	Chris Turner
Company	Repligen	Company	Silq Technologies Inc
Signature		Signature	
Date	01 Nov 2022	Date	01 Nov 2022

Comments

FSE Comments:

Customer Comments: